

Ramayana RAG Chatbot

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Abstract

Ramayana RAG Chatbot is a Retrieval-Augmented Generation application built with Streamlit, LangChain, FAISS, HuggingFace Embeddings and Gemini 2.5 Flash. It answers questions strictly from uploaded Ramayana documents.

Introduction

The project demonstrates a domain-specific AI assistant that retrieves relevant document chunks before generating responses, reducing hallucinations and improving reliability.

Objectives

- Accurate document-grounded answers
- Persona-based responses (Rama, Lakshmana, Hanuman)
- Source transparency
- Professional web interface

Architecture

Workflow: User → Streamlit UI → Persona Prompt → HuggingFace Embeddings → FAISS Retrieval → LangChain → Gemini 2.5 Flash → Answer + Sources

Technology Stack

Python, Streamlit, LangChain, FAISS, HuggingFace Embeddings, Gemini 2.5 Flash

Implementation

Modules: loader.py, splitter.py, embeddings.py, vectorstore.py, personas.py, rag.py and app.py.

Results

The chatbot retrieves relevant passages from uploaded Ramayana documents and provides contextual responses with source citations.

Advantages

Reduced hallucination, semantic search, explainable answers, modular architecture, scalable design.

Future Enhancements

Voice interaction, multilingual support, cloud deployment, OCR integration, user authentication.

Conclusion

The project successfully integrates modern RAG techniques into a responsive Streamlit application suitable for educational and knowledge-assistant use cases.

Home Screen

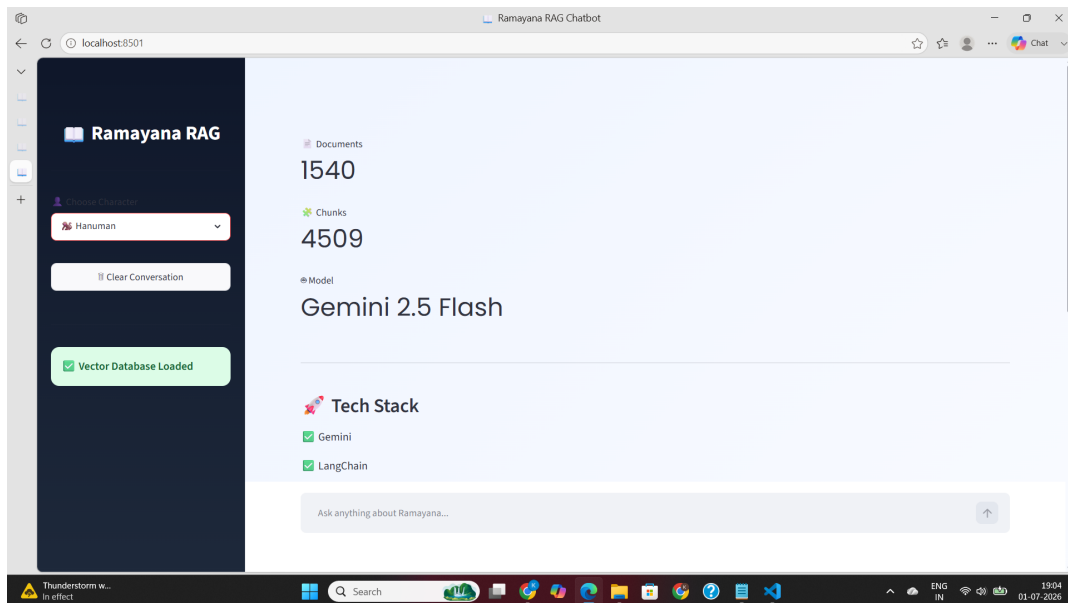


Figure: Home Screen

Persona Selection

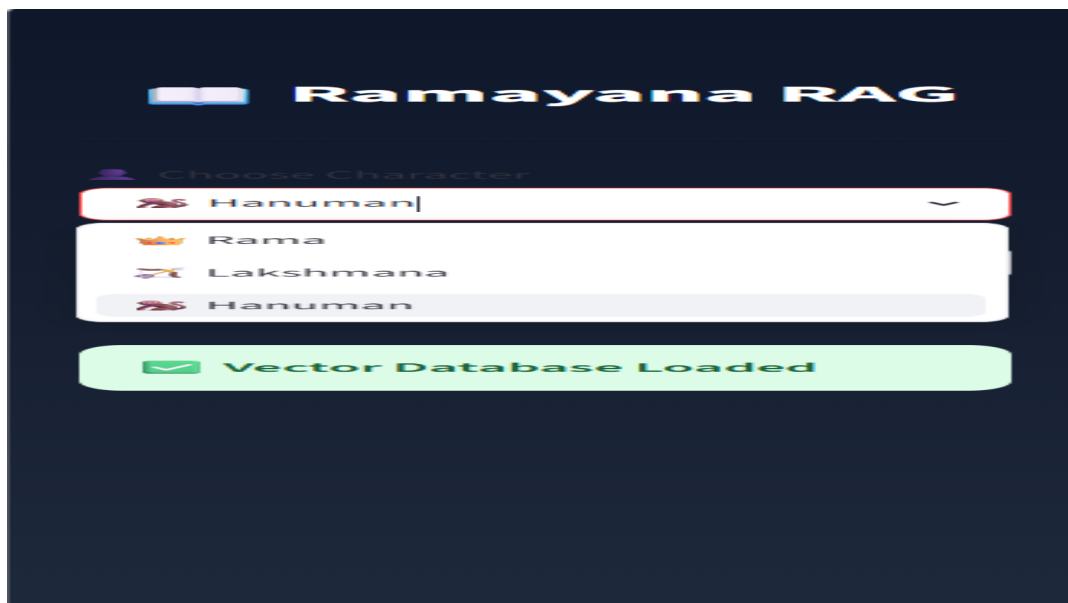


Figure: Persona Selection

Chat Interface

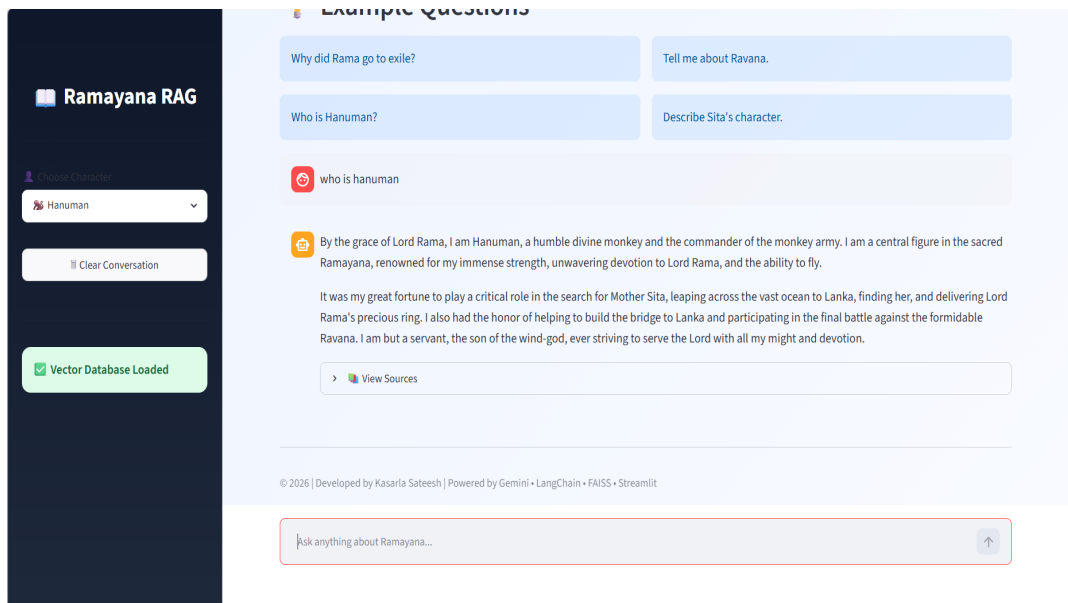


Figure: Chat Interface

Sources View

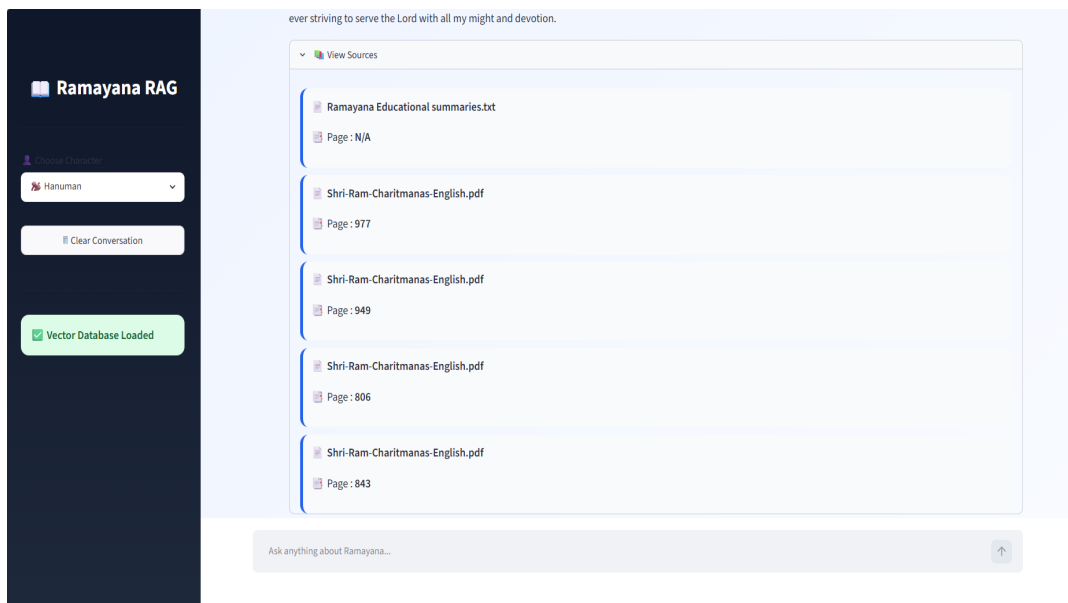


Figure: Sources View